

O(n) Time Complexity

O(n) means the time taken by an algorithm grows linearly with input size (n).

An algorithm is O(n) if it processes each element of input once

Example:-

```
my_list = [10, 20, 30, 40, 50]
```

```
for i in my_list:
    print(i)
```

Common O(n) Operations in Python:-

1. Traversing a list

```
for i in my_list:
    print(i)
```

2. Searching in a list

```
def find_element(my_list, target):
    for x in my_list:
        if x == target:
            return True
    return False
```

3. Counting elements

```
count = 0
for x in my_list:
    count += 1
```

4. String traversal

```
s = "hello"

for ch in s:
    print(ch)
```

Membership in list

```
if 10 in my_list:
    print("Found")
```

```
n = 10    →    10    steps
n = 100   →    100   steps
```

$n = 1000 \rightarrow 1000$ steps

Variations of $O(n)$

1. Single loop $\rightarrow O(n)$

```
for i in range(n):  
    pass
```

2. Multiple loops (separate) $\rightarrow$ still $O(n)$

```
for i in range(n):  
    pass
```

```
for j in range(n):  
    pass
```

3. Nested loop $\rightarrow O(n^2)$

```
for i in range(n):  
    for j in range(n):  
        pass
```